ANALYSIS OF INDIAN MUTUAL FUND DATA USING PYTHON

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Abstract

This project focuses on analysing Indian mutual fund data obtained from the India Data Portal, based on the AMFI Monthly Report, covering the period from 2019 to 2024. The main objective is to clean and preprocess the dataset, perform statistical analysis, identify trends and relationships among important financial variables, and present the findings through suitable visualisations.

The dataset contains information on mutual fund scheme types, scheme categories, number of schemes, investor folios, fund mobilised, repurchase and redemption, net inflow/outflow, and Net Assets Under Management (Net AUM). Python was used in Google Colab for data cleaning, preprocessing, exploratory data analysis, statistical analysis, and visualisation. Pandas and NumPy supported data handling and numerical operations, while Matplotlib, Seaborn, and Plotly were used to create charts and graphs. Data-cleaning steps included handling missing values, removing duplicate records, correcting data types, reviewing incorrect values, and detecting potential outliers.

The analysis showed that Open Ended Schemes had the highest number of records, while Income/Debt Oriented Schemes were the most frequent scheme category. Average Net AUM increased consistently from 2019 to 2024, reaching its highest level in 2024. Fund Mobilized and Repurchase/Redemption showed a very strong positive correlation, while average Net Inflow/Outflow fluctuated during the study period. Visualisations helped illustrate the distribution of variables, yearly trends, and relationships between financial indicators.

Overall, the project demonstrates how Python-based data analysis can be used to examine mutual fund data, identify meaningful patterns, and communicate findings effectively.

Introduction

Mutual funds are an important investment option in the Indian financial market. They collect money from investors and invest it in different financial instruments according to their investment objectives.

This project focuses on analysing Indian mutual fund data obtained from the India Data Portal. The dataset contains information about the number of mutual fund schemes, investor folios, fund mobilisation, repurchase and redemption, net inflow or outflow, and net assets under management (AUM).

Python was used to perform data cleaning, preprocessing, exploratory data analysis, statistical analysis, and data visualisation. The analysis aims to identify patterns and trends in mutual fund data and present the findings through suitable charts and graphs.

Problem Statement

The AMFI mutual fund dataset contains information about scheme types, scheme categories, fund mobilization, repurchase/redemption, net inflow/outflow, and Net Assets Under Management (Net AUM). Analysing this data manually can make it difficult to identify trends, patterns, and relationships among financial variables. Therefore, this project aims to use Python to clean the dataset, perform statistical analysis, and create visualisations to understand mutual fund trends from 2019 to 2024.

Project Objectives

The main objectives of this project are:

1. To analyse Indian mutual fund data obtained from the India Data Portal.
2. To identify and handle missing values, duplicate records, data-format issues, and other data quality problems using Python.
3. To perform exploratory data analysis (EDA) using descriptive statistics and suitable analytical techniques.
4. To analyse mutual fund scheme types, scheme categories, and numerical variables.
5. To examine trends in Net Assets Under Management (Net AUM) and net inflow or outflow over time.
6. To study relationships between numerical variables using correlation analysis.
7. To visualise the data using appropriate charts and graphs.
8. To summarise the key findings and draw conclusions based on the analysis.

Dataset Description

The dataset used for this project is based on the AMFI Monthly Report obtained from the India Data Portal. AMFI stands for the Association of Mutual Funds in India.

The dataset contains information about Indian mutual fund schemes, including scheme type, scheme category, number of schemes, number of investor folios, fund mobilised, repurchase and redemption, net inflow or outflow, and Net Assets Under Management (Net AUM).

The dataset was analysed using Python in Google Colab. Pandas and NumPy were used for data handling and preprocessing, while Matplotlib, Seaborn and Plotly were used to create visualisations.

For learning and data-cleaning practice, certain data quality issues were introduced into a working copy of the dataset. These issues were then examined and handled during the data-cleaning process using Python.

The notebook initially loaded 3,124 records and 14 columns. After removing 9 duplicate rows, 3,115 records remained for analysis. The dates represented in the cleaned notebook range from April 2019 to July 2024.

Attribute	description
Source portal	India Data Portal
Original data source	Association of Mutual Funds in India (AMFI)
Initial records	3,124
Duplicate rows removed	9
Records after duplicate removal	3,115
Original columns	14
Additional derived columns	Year and month
Periods shown in notebook outputs	April 2019 – July 2024

Source URL : IDP | Mutual Funds Report

Main Dataset Variables

Variable Group	Columns
Identification & Date	record_id, date
Scheme details	scheme_name, scheme_type, scheme_category
Scheme and folio counts	no_of_schemes, no_of_folios
Fund flow indicators	fund_mobilized, repurchase_redemption, net_infow_outflow
Asset indicators	net_aum, avg_net_aum
Segregated portfolio indicators	number_of_segregated_portfolios, net_aum_in_segregated_portfolios
Derived time variables	Year, month

Tools Used

The following tools and libraries were used to perform data cleaning, exploratory data analysis, and visualization in this project.

1. Python

Python was used as the primary programming language for data cleaning, preprocessing, analysis, and visualization.

2. Google Colab

Google Colab was used as the cloud-based environment to write and execute Python code.

3. Pandas

Pandas was used to load the dataset, inspect records, handle missing values, identify and remove duplicate records, and perform data manipulation.

4. NumPy

NumPy was used for numerical operations and data processing.

5. Matplotlib

Matplotlib was used to create graphs and charts to understand data distributions and trends.

6. Seaborn

Seaborn was used to create statistical visualizations, including distribution plots, boxplots, and correlation heatmaps.

7. Plotly Express

Plotly was used to create interactive visualizations for exploring the dataset, including a line graph and correlation heatmap.

Data Cleaning and Preprocessing

Data cleaning and preprocessing were performed using Python in Google Colab. The dataset was inspected to identify missing values, duplicate records, incorrect values, and data format issues. The identified issues were reviewed and handled using appropriate Pandas methods to prepare the dataset for exploratory data analysis.

1.Duplicate Removal

Duplicate records were identified using the ‘duplicated()’ method . A total of 9 duplicate rows were found using ‘duplicated.sum()’and removed using `drop_duplicates()`, reducing the dataset from 3,124 to 3,115 records.

2.Handling Missing Values

Missing values in the dataset were identified using the ‘isnull().sum()’ method in Python. The inspection revealed missing values in the *fund_mobilized, repurchase_redemption, number_of_segregated_portfolios, and net_aum_in_segregated_portfolios* columns.

To handle the missing values in **fund_mobilized and repurchase_redemption**, the **median** of each respective column was used for imputation. Median imputation was selected because it is less sensitive to extreme values and helps maintain the overall distribution of the data.

For **number_of_segregated_portfolios and net_aum_in_segregated_portfolios**, the missing values were retained as NaN because there was no sufficient evidence to confirm that the missing values represented zero. This helped avoid introducing assumptions that could affect the accuracy of the analysis.

After handling the missing values, the dataset was prepared for further exploratory data analysis.

3.Data Type Correction

The data types of the dataset columns were examined using ***df.dtypes*** to ensure that each variable was stored in the appropriate format.

The **date** column was found as object type and converted into a proper **datetime** format to facilitate accurate date-based analysis.

Numerical columns were checked to ensure that they had suitable numeric data types for statistical calculations and further analysis. These corrections helped maintain data consistency and prepared the dataset for exploratory data analysis.

4.Incorrect Value Handling

Some incorrect values were intentionally introduced into the dataset for learning and practice purposes. During data cleaning, negative values were identified in the **number_of_schemes** and **net_aum** columns. These values were checked against the original dataset and corrected using the corresponding reference values.

The corrections helped improve the accuracy and consistency of the dataset. However, negative values in the **net_inflow_outflow** column were retained because they represent valid net outflows rather than data errors.

Negative values in Fund Mobilized and Repurchase/Redemption were reviewed. Since these values may follow the dataset's reporting convention, they were retained without modification unless verified as errors.

This process ensured that incorrect values were corrected while meaningful financial observations were preserved for further analysis.

4.Outlier Detection and Handling

Outlier detection was performed using the Interquartile Range (IQR) method in Python. The IQR method helps identify unusually high or low values in numerical data by calculating the difference between the third quartile (Q3) and the first quartile (Q1).

IQR = Q3 − Q1

The lower and upper limits were calculated using the following formulas:

- Lower Limit = Q1 − 1.5 × IQR
- Upper Limit = Q3 + 1.5 × IQR

Values falling below the lower limit or above the upper limit were identified as potential outliers.

The IQR method was applied to all relevant numerical columns in the dataset to examine the presence of outliers. During this process, an unusually high value was identified in the `net_aum` column. This value had been intentionally introduced for learning and practice purposes as part of the data-cleaning exercise.

The identified value was checked against the original dataset and corrected using the corresponding original value. Other potential outliers were examined carefully rather than being removed automatically, as extreme values in financial data may represent genuine observations.

The segregated portfolio columns were excluded from IQR outlier analysis because most of their values are zero or missing, which can make the IQR results misleading.

This approach helped identify unusual values, correct the intentionally introduced error, and preserve meaningful financial information for further exploratory data analysis.

6. Renaming Columns

The column names in the dataset were renamed using the `rename()` function in Python to improve readability and clarity. Abbreviated column names were replaced with more descriptive names to make the dataset easier to understand and use during analysis.

The following column names were updated:

- `id` was renamed to `record_id`.
- `no_of_schemes` were renamed to `number_of_schemes`.
- `no_of_folios` were renamed to `number_of_folios`.
- `no_of_segregated_portfolios` were renamed to `number_of_segregated_portfolios`.

The `inplace=True` parameter was used to apply these changes directly to the existing DataFrame without creating a separate DataFrame.

This renaming process improved the consistency and readability of the dataset and made subsequent data analysis and visualisation easier. Snake_case naming was maintained throughout the dataset.

7. Feature Engineering

Feature engineering was performed to extract additional information from the existing date column. Using Python, two new columns, **`year`** and **`month`**, were created from the **`date`** column.

The following code was used to extract the year and month from the date column:

```
df['year'] = df['date'].dt.year
df['month'] = df['date'].dt.month
```

The `year` column was used to analyse yearly trends, while the `month` column provided information for month-wise analysis. These additional features helped make time-based analysis and visualisation more convenient.

This step improved the dataset's suitability for exploratory data analysis and helped identify changes in mutual fund-related financial indicators over time.

Exploratory Data Analysis (EDA)

1.Statistical Analysis

Statistical analysis was performed on the AMFI mutual funds dataset using Python to understand the characteristics, distribution, and relationships within the data. Descriptive statistics were used to summarise the numerical variables, while categorical and numerical data were examined separately. Correlation analysis was also conducted to understand the relationships between selected numerical variables.

These analyses provided a statistical overview of the dataset and helped identify important patterns before proceeding to data visualisation.

1.1 Descriptive Analysis

Descriptive statistical analysis was performed using Python to summarise the numerical variables in the AMFI mutual funds dataset. The describe() function was used to calculate key statistical measures, including count, mean, standard deviation, minimum, maximum, and quartiles.

These statistical measures helped understand the average values, variation, and range of the numerical data. The summary provided an overview of the dataset and supported further analysis.

The main findings are summarised below:

- **Number of Schemes:** The mean was approximately 32.90, while the median was 22.
- **Number of Folios:** The mean was approximately 2,518,408, and the median was 521,606.
- **Fund Mobilized:** The mean was 20,265.06, compared with a median of 1,025.55.
- **Repurchase/Redemption:** The mean was 19,791.06, while the median was 880.745.
- **Net Inflow/Outflow:** The mean was 473.37, and the median was 26.85. The minimum value was −157,970.38, indicating that the dataset includes observations with substantial net outflows.
- **Net AUM:** The mean was 75,805.39, and the median was 31,617.13.

- **Average Net AUM:** The mean was 76,722.93, while the median was 31,897.35.

The differences between the means and medians of several variables indicate that their distributions may be influenced by extreme values. These observations highlight the importance of examining the data distribution and potential outliers before drawing conclusions.

Overall, descriptive statistical analysis provided a numerical overview of the dataset and helped establish a foundation for the subsequent categorical, numerical, and correlation analyses.

1.2 Categorical Data Analysis

Categorical data analysis was performed using Python to understand the distribution of mutual fund scheme types and scheme categories in the dataset. The *value_counts()* function was used to count the number of records in each category.

Scheme Type Analysis

The analysis of scheme types showed that Open Ended Schemes had the highest number of records, with 2,475. Close Ended Schemes accounted for 448 records, while Interval Schemes accounted for 192 records. This indicates that Open Ended Schemes represented the largest share of records in the dataset.

Scheme Category Analysis

The scheme category analysis showed that Income/Debt Oriented Schemes had the highest number of records, with 1,344, followed by Growth/Equity Oriented Schemes with 875 records. Other Schemes and Hybrid Schemes each had 384 records, while Solution Oriented Schemes had the lowest count, with 128 records.

Overall, categorical data analysis helped identify the distribution of mutual fund scheme types and categories within the dataset.

1.3 Numerical Data Analysis

Numerical data analysis was performed using Python to examine the numerical variables in the mutual fund dataset. The analysis focused on understanding the distribution and variation of variables such as the number of schemes, number of folios, fund mobilized, repurchase/redemption, net inflow/outflow, Net AUM, and average Net AUM.

The analysis helped understand the variations in mutual fund-related numerical variables and provided insights into the financial characteristics of the dataset. These findings supported further investigation through data visualisation and correlation analysis.

1.4 Time Series Analysis

Time series analysis was performed using Python to examine yearly variations in average Net Assets Under Management (Net AUM), net inflow/outflow, fund mobilized, and repurchase/redemption from 2019 to 2024.

The analysis revealed a consistent increase in average Net AUM throughout the study period. Average net inflow/outflow fluctuated over the years, with a significant increase in 2024. In contrast, average fund mobilized and repurchase/redemption declined until 2021 and gradually recovered in the following years, although both remained below their 2019 levels.

Overall, the analysis helped identify yearly trends and variations in important mutual fund financial variables, providing insights into changes in asset management and fund-related activities during the study period.

1.5 Correlation Analysis

Correlation analysis was performed using Python to examine the relationships between the numerical variables in the AMFI mutual funds dataset. The correlation matrix was used to identify the strength and direction of relationships between the variables.

The analysis showed a very strong positive correlation between Fund Mobilized and Repurchase/Redemption (0.9965). Net AUM and Average Net AUM also showed a very strong positive correlation (0.9968).

Number of Folios had a moderate positive correlation with Net AUM (0.6224) and Average Net AUM (0.5965). This indicates that these variables tended to increase together in the dataset.

Number of Schemes showed a weak positive correlation with Net AUM (0.2702) and Average Net AUM (0.2637). Net Inflow/Outflow had relatively weak correlations with most of the other variables.

The segregated portfolio variables were excluded from this correlation analysis because their values were largely zero or missing. The analysis was therefore focused on the seven selected numerical variables.

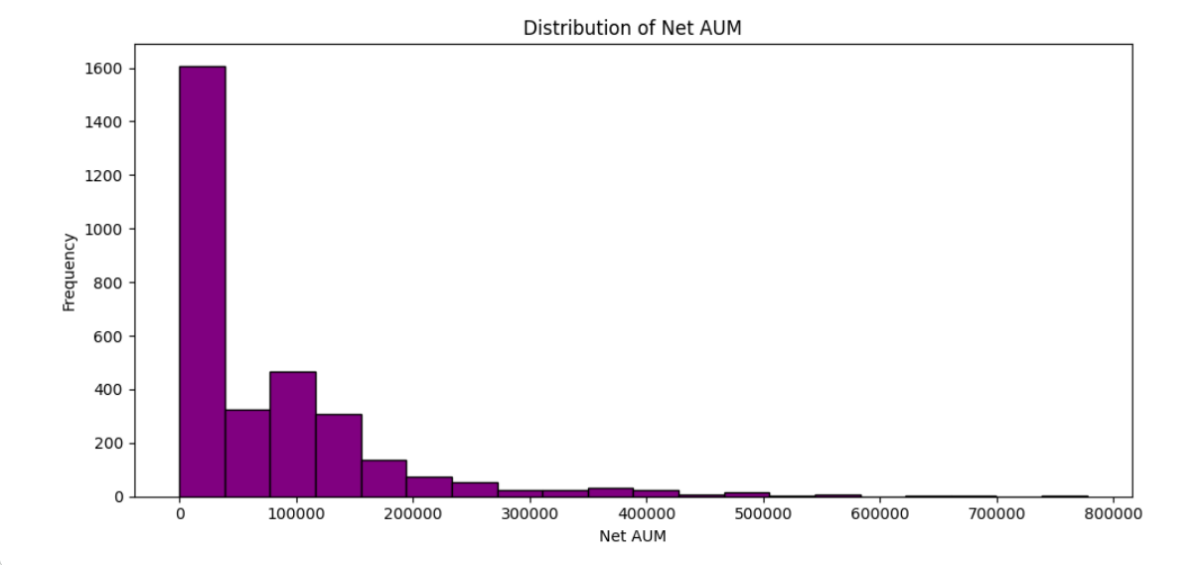
Overall, correlation analysis helped identify strong and weak relationships among the selected variables. However, correlation indicates association and does not necessarily imply causation.

Data Visualization

Data visualization was performed using Python to represent the patterns, distributions, and relationships in the mutual fund dataset through graphical representations. Different types of charts were used to understand the distribution of variables, compare financial measures across categories and years, and identify relationships between numerical variables.

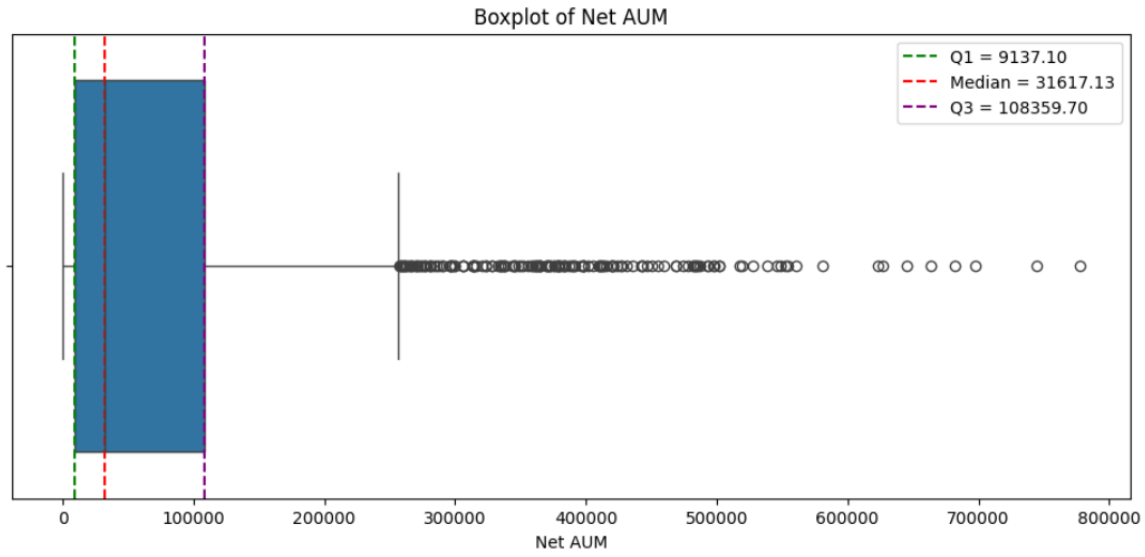
1. Histogram of Net AUM

The histogram was used to examine the distribution of Net AUM values and understand how the values are spread across different ranges.



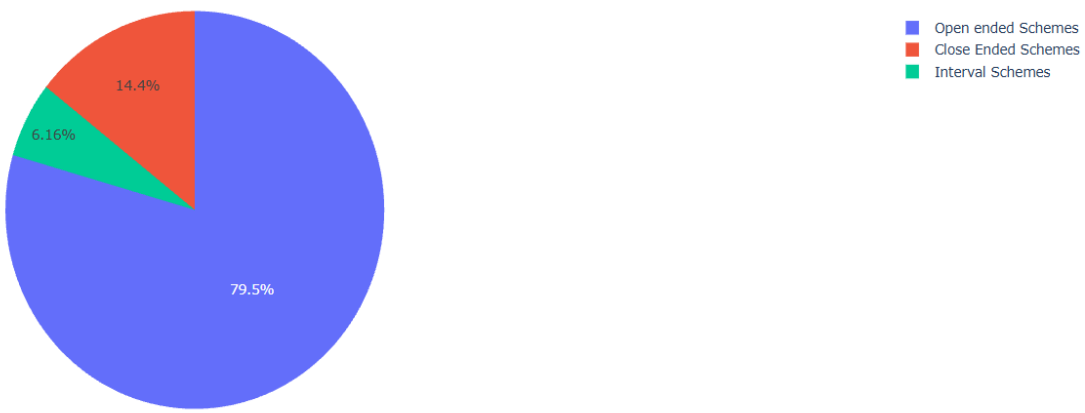
The histogram shows a right-skewed distribution of Net AUM. Most observations fall within the lower value ranges, while a smaller number of observations have substantially higher Net AUM values, forming a long right tail.

2. Net AUM Box plot

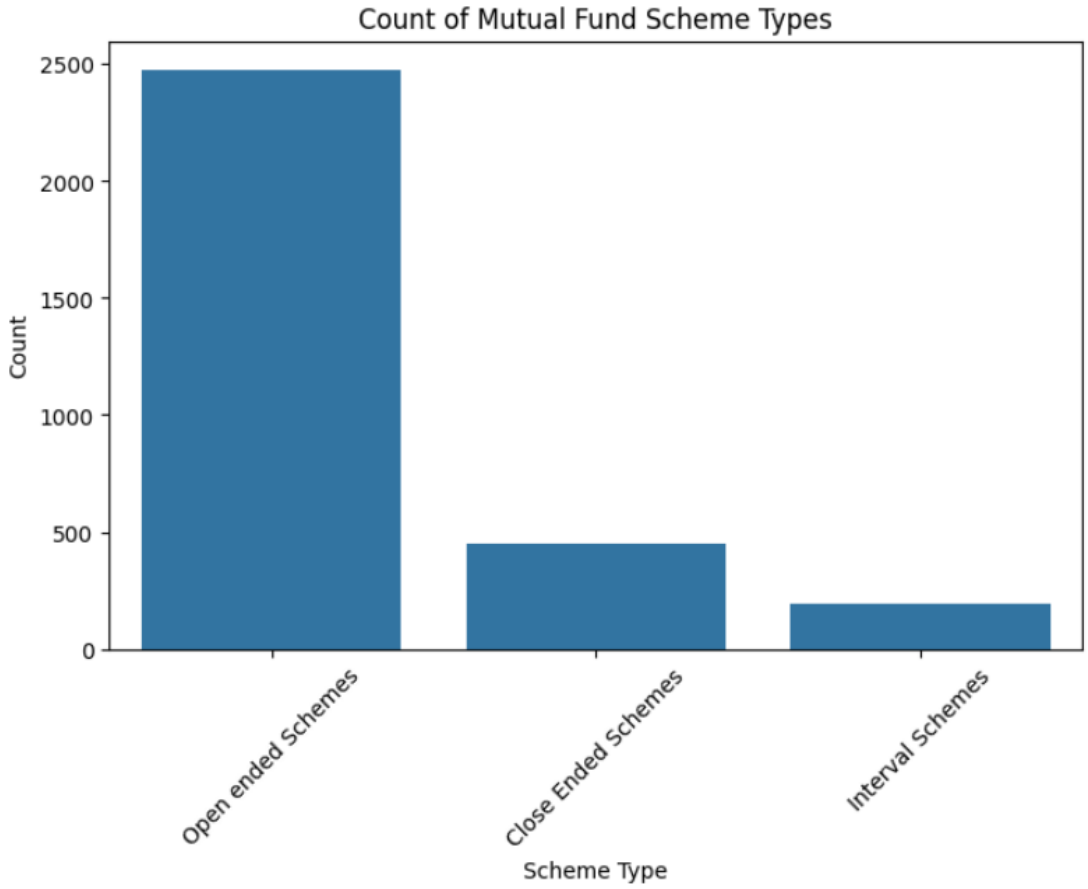


The boxplot shows the median and spread of Net AUM values. The interquartile range is wide, and several observations lie beyond the upper whisker, indicating potential high-value outliers.

3.Pie Chart: Distribution of Mutual Fund Scheme Types

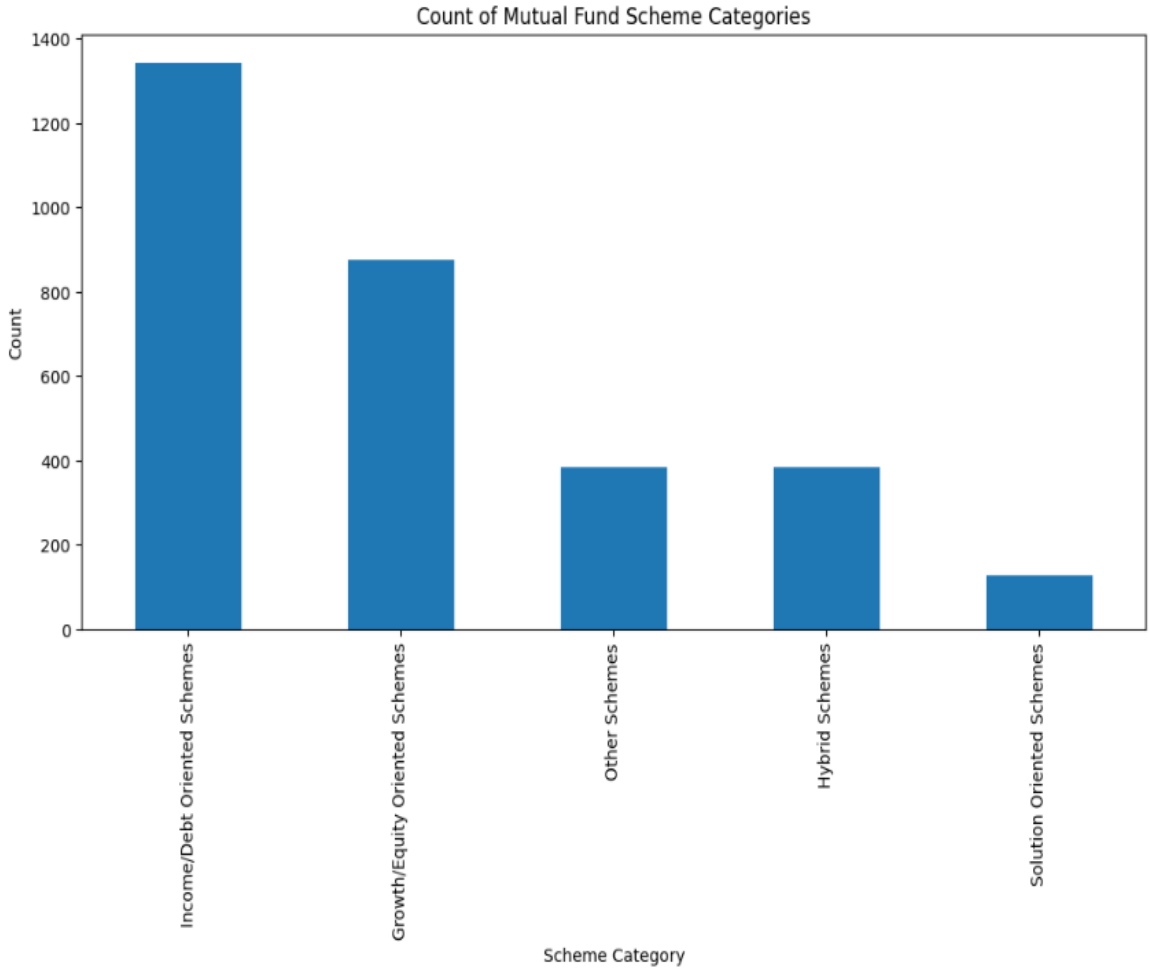


4. Scheme Type Count



The count plot shows that Open Ended Schemes account for the largest number of records, followed by Close Ended Schemes and Interval Schemes . Thus, Open Ended Schemes dominate the dataset in terms of record count.

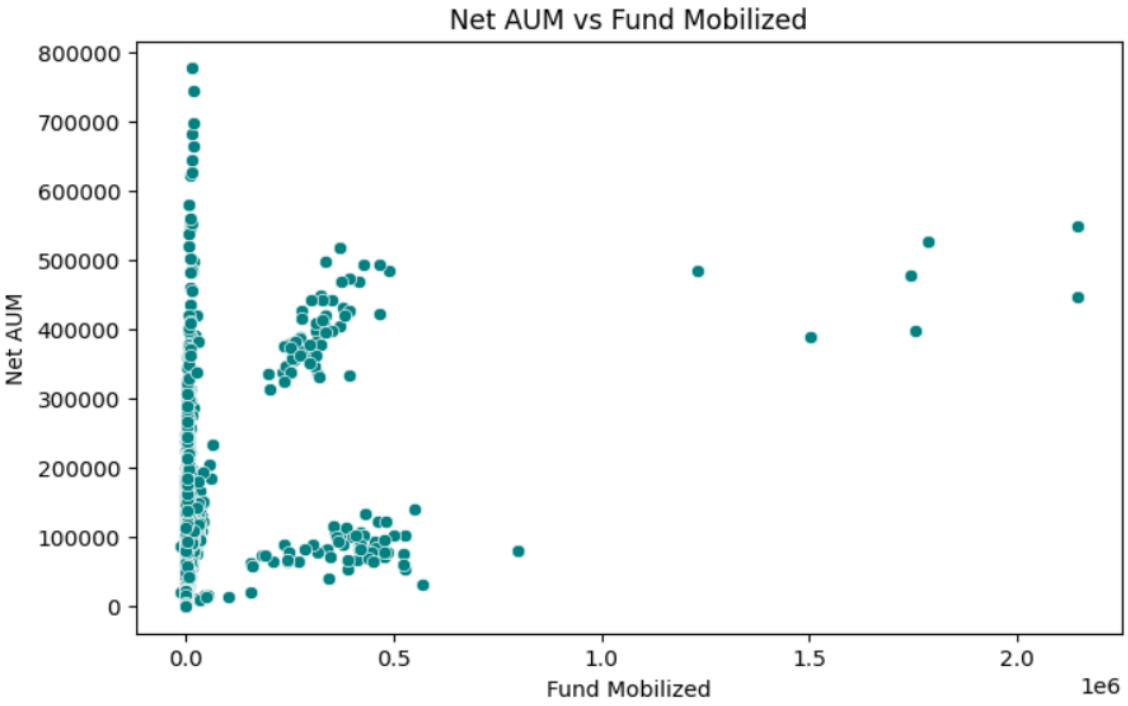
5.Scheme Category Distribution



A bar chart was created to visualize the number of records in each mutual fund scheme category. This helps compare the frequency of different scheme categories in the dataset.

The bar chart shows that Income/Debt Oriented Schemes have the highest record count, followed by Growth/Equity Oriented Schemes. Other Schemes and Hybrid Schemes each have records, while Solution Oriented Schemes have the lowest count.

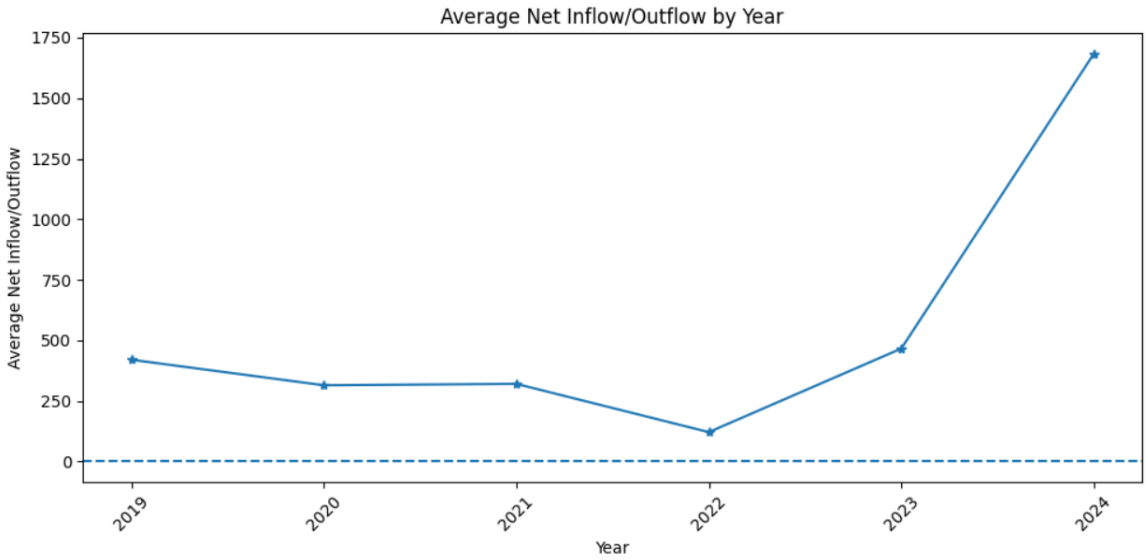
6.Net AUM vs Fund Mobilized



A scatter plot was created to examine the relationship between Fund Mobilized and Net AUM. The plot helps visualize the pattern and strength of the relationship between these two financial variables.

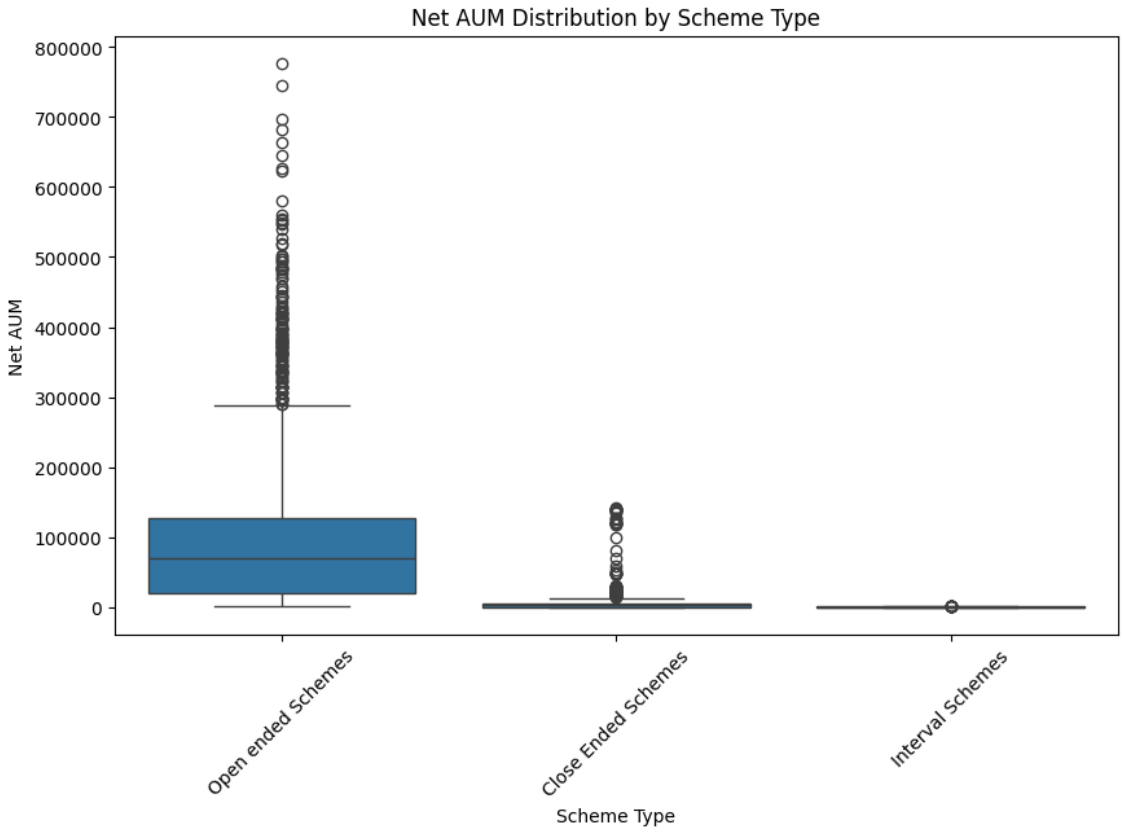
The scatter plot shows a dispersed relationship between Fund Mobilized and Net AUM, with many observations clustered at lower Fund Mobilized values and a few extreme values at the upper end. The pattern suggests that the relationship is not strongly linear in the plotted data, and the extreme observations should be interpreted carefully.

7.Average Net Inflow/Outflow by Year



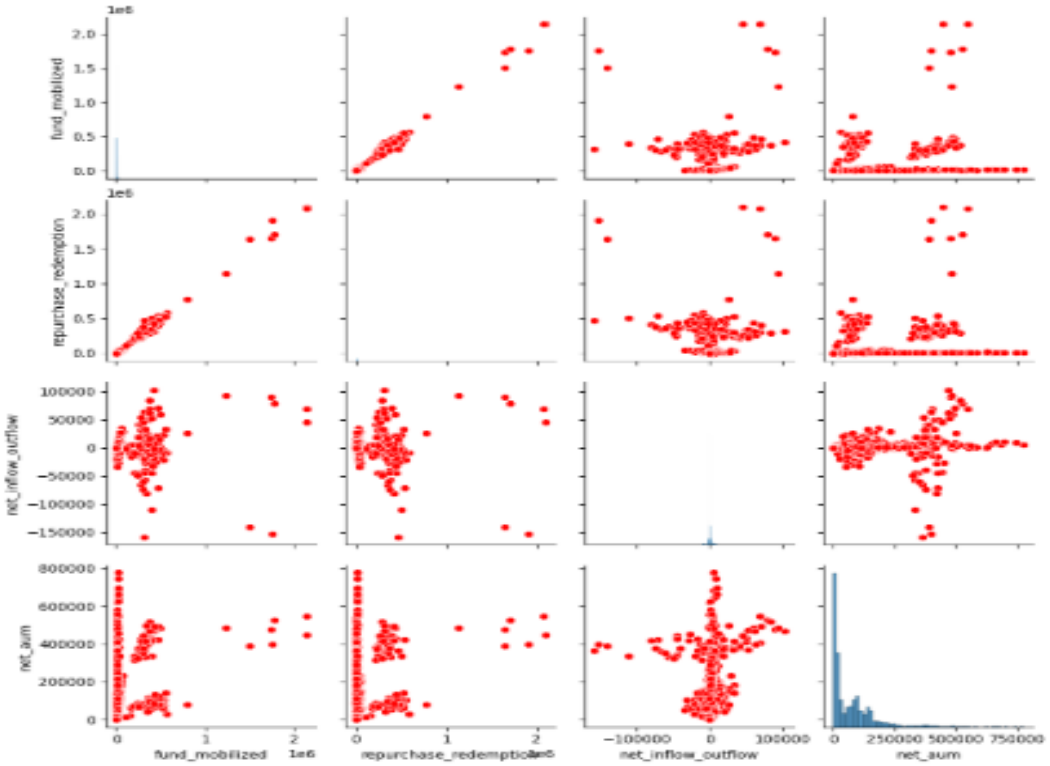
The line graph shows the yearly trend in average Net Inflow/Outflow from 2019 to 2024. The average decreased from approximately 420 in 2019 to 315 in 2020, remained nearly stable in 2021, and reached its lowest level of approximately 122 in 2022. It then increased to around 466 in 2023 and rose significantly to approximately 1,683 in 2024.

8.Scheme Type vs Net AUM — Boxplot



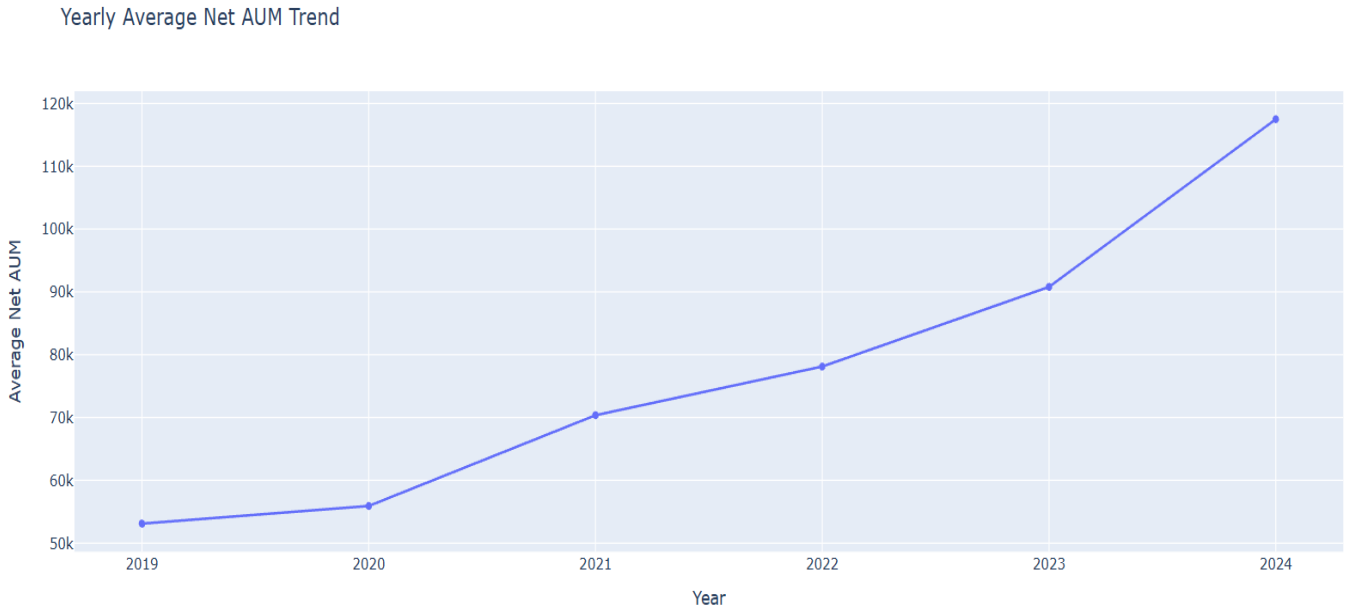
The boxplot compares the distribution of Net AUM across scheme types. Open-ended schemes show the highest median and the widest spread, along with numerous high-value outliers. Close-ended and Interval Schemes have substantially lower Net AUM distributions in comparison.

9.Pair Plot



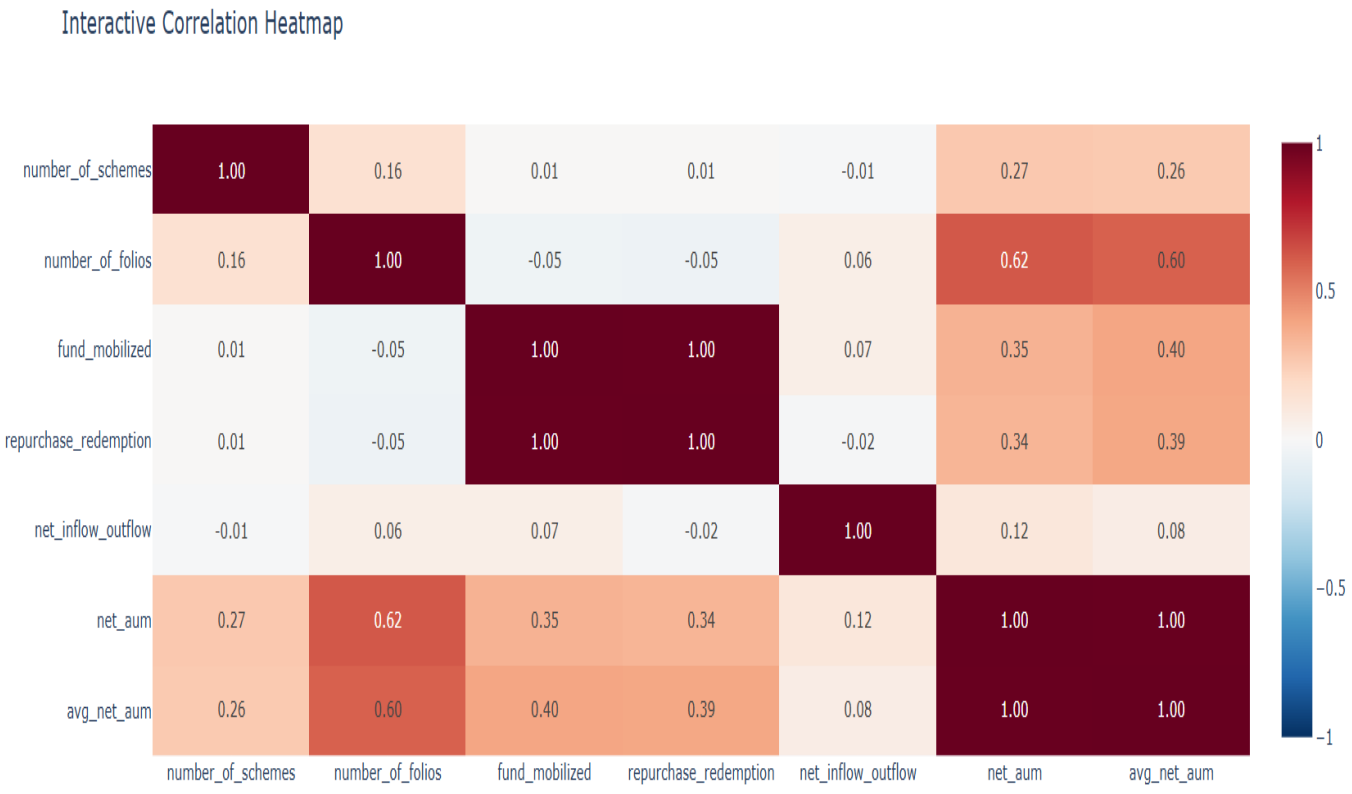
The pair plot compares the distributions and pairwise relationships of four key financial variables. Fund Mobilized and Repurchase/Redemption show a very strong positive linear association. Net AUM has a more dispersed relationship with these variables, while Net Inflow/Outflow shows a less distinct pattern. The plots help identify relationships and unusual observations, but they do not establish causation.

10.Interactive Line Graph



The interactive line graph shows the trend in yearly average Net Assets Under Management (Net AUM) from 2019 to 2024. The average Net AUM increased steadily from approximately 53,104 in 2019 to 117,486 in 2024. The highest average value was recorded in 2024. This indicates an overall upward trend in the average Net AUM during the study period.

11.Interactive Correlation Heatmap



The interactive correlation heatmap illustrates the strength and direction of linear relationships between the numerical variables in the dataset. A very strong positive correlation was observed between Net AUM and Average Net AUM (approximately 0.997), as well as between Fund Mobilized and Repurchase/Redemption (approximately 0.996). The number of folios showed a moderately strong positive correlation with Net AUM (approximately 0.622). In contrast, Net Inflow/Outflow showed a weak positive correlation with Net AUM (approximately 0.116). These results indicate associations between variables but do not establish causation.

Key Findings and Insights

- The dataset contained **3,115 records** after removing 9 duplicate rows during data cleaning.
- **Open Ended Schemes** represented the largest group by scheme-type record count, while **Income/Debt Oriented Schemes** had the highest number of records among scheme categories.
- Average Net Assets Under Management (Net AUM) showed a consistent upward trend from 2019 to 2024, reaching its highest average in 2024.
- Average Net Inflow/Outflow fluctuated during the study period and reached its highest average in 2024.
- Average Fund Mobilized and Repurchase/Redemption declined until 2021 and then showed a gradual recovery. However, their average values in 2024 remained below their 2019 levels.
- Correlation analysis revealed a **very strong positive relationship** between Fund Mobilized and Repurchase/Redemption. Net AUM and Average Net AUM also showed a positive relationship.
- The Net AUM histogram indicated a right-skewed distribution, with several high values identified for review using the IQR method. These extreme observations were reviewed, and genuine values were retained rather than automatically removed.

Overall Insight

The analysis highlighted the growth in average Net AUM between 2019 and 2024, along with yearly fluctuations in other mutual fund financial variables. The strong correlation between Fund Mobilized and Repurchase/Redemption indicated that these two variables moved closely together in the dataset. Overall, the findings provided a clearer understanding of mutual fund trends, category distributions, and relationships among key financial variables during the study period.

Recommendations

Based on the analysis of the AMFI mutual fund dataset, the following recommendations are suggested:

- **Monitor Net AUM Trends:** Mutual fund performance trends should be monitored regularly to understand changes in assets under management over time.
- **Analyse Fund Mobilization and Redemption:** Fund Mobilized and Repurchase/Redemption should be reviewed together because the analysis showed a very strong positive correlation between these variables.
- **Review Net Inflow/Outflow:** Yearly changes in net inflow/outflow should be examined to better understand variations in fund inflows and outflows.
- **Compare Scheme Types and Categories:** Comparing Net AUM across different scheme types and categories can help identify differences in their contribution to the overall dataset.
- **Improve Data Quality:** Missing values, duplicate records, inconsistent formats, and unusual observations should be carefully checked during data cleaning. Extreme values should be verified against reliable sources before any correction or removal.
- **Use Regular Data Visualisation:** Histograms, boxplots, trend graphs, and correlation heatmaps can help identify patterns and support clearer interpretation of mutual fund data.

These recommendations can support better data quality, trend monitoring, and interpretation of mutual fund statistics. The findings describe patterns in the dataset and should not be treated as direct investment advice.

Conclusion

This project applied Python-based data cleaning, exploratory data analysis and visualisation to an AMFI mutual fund dataset. The analysis identified the most frequent scheme types and categories, summarised important numerical variables, examined year-wise changes in fund flows and Net AUM, and explored correlations among key financial indicators. The results show an overall increase in average Net AUM over the period represented in the notebook, alongside changes in mobilisation, redemption and net inflow/outflow. The visualisations made these patterns easier to interpret. The project also demonstrates the importance of reviewing missing values, duplicates, data types and outliers carefully before drawing conclusions.

Limitations and Future Scope

Limitations

- The analysis identifies patterns and relationships in the dataset but does not establish cause-and-effect relationships.
- The segregated-portfolio columns were excluded from the main correlation and IQR analyses because they contained predominantly zero or missing values.
- Median imputation was used to handle missing values in Fund Mobilized and Repurchase/Redemption. This may have influenced the summary statistics and analysis results.
- Some incorrect values, duplicates, and a Net AUM outlier were intentionally introduced for learning purposes. Corrections were made during data cleaning with reference to the original source data, as documented in the notebook.
- The dataset covers the period from 2019 to 2024 only. Therefore, the analysis does not reflect trends beyond 2024, and the findings may not represent more recent developments in the mutual fund sector.
- The measurement units and date information should be verified against the original source documentation to ensure accurate interpretation of the results.

Future Scope

- The analysis can be extended by including more recent AMFI data to examine mutual fund trends over a longer period.
- Further analysis can explore changes in scheme types and categories and compare their Net AUM over time.
- Additional statistical techniques and visualisations can be used to investigate relationships among financial variables in greater detail.
- Future projects can incorporate forecasting techniques to study possible trends in Net AUM and fund flows, while recognising that forecasts are subject to uncertainty.
- An interactive dashboard using tools such as Power BI can be developed to present key findings and trends more effectively.

References

1. India Data Portal – AMFI Monthly Report: https://indiadataportal.com/p/investments/r/amfi-amfi_monthly_report-in-mn-aaa
2. Google Colab: https://colab.research.google.com/